Table 2. EFFECT OF ANTI-INSULIN SERUM ON THE RESPONSE TO RE-FEEDING OF HEPATIC UDPG-GLYCOGEN GLUCOSYLTRANSFERASE ACTIVITY, GLYCOGEN CONCENTRATION, AND BLOOD GLUCOSE CONCENTRATION IN RATS FASTED FOR 72 H

| Group | No. of animals | Glucosyltransferase activity (μmoles glucose/g protein/h) −G-6-P | +G-6-P | Glycogen conc. (mg/g) | Blood glucose (mg/100 ml.) |
|---|---|---|---|---|---|
| Control | 3 | 71 ± 8·8 | 480 ± 13 | 2·2 ± 0·6 | 76 ± 2·2 |
| Re-fed | 3 | 480 ± 107 | 700 ± 107 | 28 ± 1·4 | 113 ± 12 |
| Re-fed + AIGPS | 3 | 118 ± 14 | 450 ± 21 | 6·9 ± 1·0 | 275 ± 3·5 |
| Re-fed + NGPS | 3 | 430 ± 117 | 680 ± 56 | 28 ± 2·2 | 109 ± 7·8 |

Female Sprague–Dawley rats, fasted for 72 h, were divided into four equal groups. Two groups were given intraperitoneal injections of either 1·5 ml. of normal guinea-pig serum (NGPS) or 1·5 ml. of anti-insulin guinea-pig serum (AIGPS) 20 min before re-feeding, and an additional 1·0 ml. of the same serum 90 min after re-feeding. Control animals were not given food or injections. The animals were killed 3·5 h after re-feeding and the livers perfused with cold 0·3 M sucrose, 0·005 M EDTA, *p*H 7·4, via the portal vein before removal. The preparation of homogenates, assay and other analytical procedures have been described (refs. 7 and 8).

response cannot be excluded. Further investigation is required to clarify these points.

It has been reported recently that synthesis of glycogen in muscle is regulated by a mechanism which can be influenced by insulin and epinephrine, and also by muscle contraction[13]. The central feature of this mechanism is the circumvention of the normal allosteric mechanism for regulation of muscle glucosyltransferase, that is, glucose-6-P dependency, by the conversion of this enzyme to a glucose-6-P independent (I) form by means of a reversible dephosphorylation of the enzyme protein reminiscent of the transformations of muscle phosphorylase[14]. It is not clear at present whether this kind of mechanism, in addition to those discussed earlier in this report, participates in the regulation of glycogen synthesis in the liver, for, in contrast to the more stable muscle glucosyltransferase, the hepatic enzyme readily loses a large proportion of its glucose-6-P dependency on mild heating at 37° in the absence of glucose-6-P, and recovers it on further incubation in the presence of glucose-6-P and certain salts[15]. It is possible that this lability of the allosteric site of hepatic glucosyltransferase may explain the greater glucose-6-P independence of newly synthesized glucosyltransferase (note glucosyltransferase activity in absence of glucose-6-P in Figs. 1 and 2, and in Table 2).

If space allowed, many examples could be cited where selective control of the rate of synthesis of enzymes has been demonstrated by manipulation of the endocrine or nutritional state of animals. The inhibition of such enzyme adaptations by actinomycin *D* is regarded as presumptive evidence that the step in enzyme formation which is subject to control is that of transcription of information from DNA to messenger RNA. While control of transcription of a rapidly turned-over messenger RNA would seem to provide the necessary precision and flexibility of control, and is consistent with hypotheses of bacterial enzyme regulation[16], many problems, which are not yet resolved, have arisen in the application of this simple concept to enzyme synthesis in more complex organisms. For example, there is still considerable uncertainty regarding the stability of the total messenger RNA fraction in tissues such as mammalian liver[17], and, moreover, there are now indications that messenger RNA function may also be subject to regulation[18]. In the case of hormone-dependent acceleration of enzyme synthesis in mammalian tissues there is also considerable doubt that a hormone acts as an inducer or derepressor as defined in the more simple bacterial model, for although the hormone is essential to the process, it appears to be only one component among several which summate to modulate the rate of production of the enzyme protein. This is exemplified by the glucosyltransferase transitions which have been discussed here. The level of this enzyme is not significantly reduced by the insulin deficiency of severe diabetes, and it seems to require the presence of some unknown metabolite(s) in order to be induced by insulin or, conversely, to require insulin for its induction by some metabolite(s). Moreover, in contrast to the adult animal, accelerated synthesis of hepatic glucosyltransferase during the latter stages of gestation of the rat fœtus is dependent on the presence of cortisol rather than insulin[19].

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# PROGNOSIS FOR INTERRUPTION OF MALARIA TRANSMISSION THROUGH ASSESSMENT OF THE MOSQUITO'S VECTORIAL CAPACITY

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WHEREVER malaria eradication relies on vector control, efforts should be made to assess the degree of control achieved and its effectiveness in producing a downward trend in malaria transmission.

I propose that 'vector control' should refer, in this context, to reduction of the mosquito population's vectorial capacity. The vectorial capacity of a population is defined as the average number of inoculations with a

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specified parasite, originating from one case of malaria in unit time, that the population would distribute to man if all the vector females biting the case became infected. This definition, from Garrett-Jones and Grab (in the press), is an attempt to give more precision to one published by me in an earlier paper[1]. Vector control is epidemiologically effective when the vectorial capacity is such that a declining rate of transmission, leading to its interruption, can be predicted.

Vectorial capacity in this sense may be assessed as the product of three parameters: (*a*) *the man-biting rate*, representing the incidence of biting-contact between the mosquito and the people in terms of number of bites per person per night; this indicates the average number of vector females liable to become infected per case per day; (*b*) *the expectation of infective life*, or days of infective life per mosquito infected with the given parasite species; (*c*) *the man-biting habit*, or bites on man per day per individual female.

Table 1 illustrates degrees of control of a vector population having a natural man-biting rate of 10, man-biting habit of 0·25, female daily survival rate of 0·9 and transmitting *P. falciparum* with 12-day sporogony.

Table 1

| | Proportion surviving per day | Man-biting rate (bites) | Expectation of infective life (days) | Man-biting habit (bites) | Vectorial capacity (inoculations) |
|---|---|---|---|---|---|
| Pre-attack | 0·9 | 10·0 | 2·68 | 0·25 | 6·70 |
| Attack: | | | | | |
| (*a*) | 0·8 | 4·72 | 0·308 | 0·25 | 0·364 |
| (*b*) | 0·7 | 2·96 | 0·0388 | 0·25 | 0·0287 |
| (*c*) | 0·7 | 1·48 | 0·0388 | 0·125 | 0·0072 |

Three possible situations soon after spraying are shown, that is, before there has been time for output from the breeding-places to be affected. In lines (*a*) and (*b*) the indicated reductions in daily survival entail the reductions shown in the man-biting rate and expectation of infective life (life beyond 12 days from time of infection); hence the much sharper reductions in vectorial capacity. Line (*c*) postulates a halved man-biting habit (by deviation to animals, as may be common in sprayed areas) superimposed on reduced longevity. This, by halving also the man-biting rate, reduces vectorial capacity by a further 75 per cent. (From any given birth-rate the density of a population with constant mortality through life varies as its expectation of life. Hence, a mosquito's man-biting rate may be expected to vary directly as the female's expectation of life. For example, when the proportion surviving through one day is reduced from 0·9 to 0·8 the expectation of life is thereby reduced from 9·49 to 4·48 days[1] (Garrett-Jones and Grab, *Bulletin of the World Health Organization*, in the press), with the immediate result that the density and the man-biting rate are reduced proportionately. The relevant mathematical formulæ are given by Macdonald[3] and by Garrett-Jones[1].)

In terms of epidemiological effectiveness, in situation (*c*) the unit source (that is, the infective malaria case) could give rise to about one infective bite per 140 days (1/0·0072), which is probably below the critical malaria reproduction level even where most cases remain untreated. In (*b*) there could be one infective bite per case per 35 days, long enough to detect and treat most cases in a malaria-eradication programme. Ineffective vector control is represented by (*a*), where the source could give rise to an infective bite about once in three days.

In any given environment and situation (such as one of those postulated here), the vectorial capacity for *P. vivax* of a mosquito population fully receptive to infection is necessarily greater—and its degree of reduction by vector control is less—than for *P. falciparum*, owing to the quicker sporogony of *P. vivax*.

Where two or more vector species occur, their joint potential for transmitting a given parasite is equal to the sum of their vectorial capacities. This potential rate of transmission, which has been called the parasite's 'daily reproduction rate'[1], refers to the potential number of infections distributed *per case per day*, in contradistinction to the basic reproduction rate which refers to the potential number distributed *per case*[3,4]. (An example of how the basic reproduction rate may be calculated is cited from Macdonald[3] by Pampana[5]. The assumed situation is that postulated in line 1 of Table 1, except that the vector's man-biting habit is given as 0·5. In those circumstances the basic reproduction rate, from a case infective for 80 days, is shown to be about 1,120—thus: $10 \times 2{\cdot}68 \times 0{\cdot}5 \times 80 = 1{,}072$.)

Vectorial capacity is a characteristic of a mosquito population related to the ecological conditions, and may be assessed irrespective of whether any cases of malaria are present at the given place and time. It thus differs radically from the sporozoite rate, the 'average infective density', or any other index of vector efficiency as defined by the World Health Organization[4]. Obviously, in any malarious area the vector's vectorial capacity will be subject to natural variations (according to relative density, longevity and man-biting habit); before the start of an attack on malaria these variations might be assessed to reveal, not the month of maximal transmission, but the month of steepest upward trend in transmission (which must happen somewhat earlier in the season). Then, when the vector is subjected to a fixed régime of control, assessments of its vectorial capacity might indicate whether the trend of transmission is consistently downwards or takes an upward turn in certain localities and in certain months. Finally, when no known malaria cases remain, knowledge of the mosquito's vectorial capacity may give warning of where and when there is danger of transmission becoming re-established, in the event of the importation of one or more cases.

It is submitted, therefore, that the probability of achieving interruption of transmission in a given area by means of the anti-mosquito measures in force, and, afterwards, the risk of resumption of transmission in that area, should be judged by the same type of criterion—the daily reproduction rate, assessed through the mosquito's vectorial capacity. The criteria for assessing the probability of interruption and of resumption of transmission were discussed by the WHO Expert Committee on Malaria at its eleventh meeting (Geneva, 1964). It is anticipated that these matters will be dealt with in the Committee's report (World Health Organization, in the press).) The resulting advantage in the continuity of entomological approach to assessment, in the successive phases of eradication, needs no stressing.

The problems of assessing vectorial capacity in field practice are essentially ones of sampling and of scale. Direct biting-catches on human baits probably offer the best hope of estimating the average incidence of contact between the vector species and the people. But the great variability of that contact would seem to dictate that the sampling be repeated on several nights per month, using a large group of human baits, extending the catches through the whole of the mosquito's biting cycle and to those stations (indoor and/or outdoor) where the people are to be found at each period. These problems are discussed by me in an unpublished WHO document (*WHO/Mal/450*).

Estimation of the expectation of infective life involves observations on the proportion of a sample surviving in Nature over a known interval of days. As the technique of counting the dilatations in the ovariole is still too difficult for routine application to tropical vectors, it is more usual to determine the proportion of parous females. The technique of Detinova[6] is gaining increasingly wide application. However, it is important to sample the population in such a way that the mean difference in age between the nullipars and the primipars in the sample is ascertainable. If females caught biting man are used, as advocated by Hamon[7] and other workers, it often appears justifiable in warm climates

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to take this interval as equivalent to three days. Samples thus caught in ovarian stages I and II are rapidly dissected to determine the proportion with uncoiled tracheolar skeins on the ovaries. The computation of the expectation of infective life from these data is discussed by Garrett-Jones and Grab (in the press).

Finally, the man-biting habit is itself compounded of the frequency of feeding and the human blood index of the mosquito[3]. The former has often been equated with the gonotrophic cycle, but this appears to need further study in many areas. Accurate assessment of the proportion of blood meals containing human blood in a given sample is carried out by the Lister Institute, England, in collaboration with the World Health Organization[5]. On the other hand, the unbiased sampling of blood meals in the field presents rather formidable difficulties, as I have already shown[1]. The samples should reflect the day-time distribution of the blood-fed females, but often we do not know the distribution of the types of resting place, let alone that of the mosquitoes. In sprayed areas the problem is still worse, since indoor-feeding mosquitoes may die before the daytime samples are collected, and then no form of diurnal sampling of survivors can fully represent the feeding population of the previous night. This problem merits urgent investigation in view of the probability that DDT and other insecticides may reduce vectorial capacity by diverting the mosquitoes to animals as well as by killing them.

In various malaria-eradication projects increasing attention is being given to the measurement of one or more of the factors determining vectorial capacity. Sustained attempts to evaluate all the factors together have been made at a few places, for example, in the Bobo-Dioulasso District of Haute Volta, where spraying with DDT failed to interrupt transmission of *P. falciparum* by *A. gambiae* and several associated vectors[2]. Yet it is true to say that nowhere, as yet, has an investigator combined his results with the view of estimating (without reliance on the sporozoite rate) the vectorial capacity of the mosquito populations under study. Neither has the daily reproduction rate of malaria—referring to the potential incidence of new infections per existing case per day—been considered in its bearing on the prognosis for interruption of transmission by the use of residual insecticides.

Despite the difficulties alluded to, it should be a primary concern of malaria entomologists to assess vectorial capacity and evaluate the resulting epidemiological trend. It is an assessment deserving priority as an operational criterion of insecticidal attack. Moreover, the problems involved constitute as strong a scientific challenge as may be encountered in most laboratory research. This fact, considered together with the epidemiological importance of the assessment, justifies the recruitment by malaria-eradication services of well-qualified scientists, and requires that they be carefully trained for these specialized entomological duties.


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# THE CHINESE WATER-BALANCE ESCAPEMENT

BURSTALL, Lansdale and Elliott[1] described the third of a laboratory series of water-wheel models of the mechanical escapement in Su Sung's clock, and expressed their belief that it represented the system used in the Chinese original of the eleventh century. From consideration of the nature and performance of the original I cannot agree that the representation is adequate.

Needham *et al.*[2,3] showed that the escapement wheel of A.D. 1088, 11 ft. in diameter, drove an assembly of heavy time-indicating and astronomical machinery. This comprised two vertical shafts about 15 ft. and 20 ft. long, six or eight pairs of gear-wheels including two or three separate solar-to-sidereal time-conversion trains, six jack-wheels over 6 ft. diameter carrying calendar-work and 150 time-indicating jacks, a 4½ ft. diameter celestial globe, and an observational armillary sphere with a 6 ft. diameter bronze rotating component which itself weighed over 150 lb.

It has been shown elsewhere[4] that there was one escapement action every 24 sec. At each action static friction had to be overcome, and the escapement wheel and machinery accelerated from rest; work had then to be done in movement against friction and the resistance of the striking mechanisms and other jack-work. Motive power was supplied by a regular flow of water at not more than 25 lb./min entering the buckets of the water wheel near its horizontal diameter. Even if a conversion efficiency of 100 per cent could be achieved in the water-wheel, an average power of only about 1/25 h.p.—28 watts—would be available for work. Yet the first laboratory model of Burstall *et al.* was designed to spill the water from each one of its eight fixed buckets as soon as the wheel moved[5,6]; and although not shown in their Fig. 1 a similar arrangement was evidently relied on in the third model to allow the 36-compartment wheel to reciprocate as described[1].

My assumption that in reality the 36 buckets emptied lower down the wheel is supported by the fact that a receiving vessel 11·4 ft. long[12] was provided to catch the water, and was positioned to extend below the entire lower half of the wheel.

The timekeeping of the water-balance escapement was such that Su Sung[7] was able in his astronomical chapter[8] to give data of the Sun's apparent position, in relation to the quasi-sidereal rotating co-ordinate system of the armillary sphere, with a precision of ±1/16th of a Chinese degree of arc. This implies the maintenance, at least over the period between one sunset, sunrise, or noon observation and the next, of a time-keeping accuracy within ± 15 sec, which is consistent with the choice of a 24-sec time unit for the escapement, and with Su Sung's interest in the 'hurrying and dawdling' of the Sun in its annual circuit.

Good timekeeping depended on accurate determination of the time unit of 24 sec by division of the regular flow of water into equal portions of not more than 10 lb. successively in each bucket. The weighing operation must surely have been done under excellent conditions, but the models of Burstall *et al.* have provided for measuring only the variable small net force likely, under real conditions, to be exerted on an external balance-lever by a rigid water-wheel, while in the act of initial movement against the static friction of all the machinery. It seems to me that even if the original escapement could operate at all on this system, it would have been unlikely to keep good time.

A practical reconstruction, avoiding the difficulties of a rigid water-wheel and providing a truly chrono-

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